

William Zhang

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Education

University of Waterloo, Computer Engineering

Waterloo, Ontario

- Current coursework includes Operating Systems, Control Systems, Digital Hardware, and Linear Algebra 2.

Projects

C Compiler

Learning compiler construction by building a C compiler in Rust.

- Systems programming
- Rust

Tiny Tapeout AI Chip

Building a custom AI chip on Tiny Tapeout in Verilog and Python with a custom PyTorch compiler.

- Digital hardware
- Verilog
- Python
- PyTorch compiler tooling

Algorithmic Trader

Developing an algorithmic stock and options trader using the CVXPY Python library.

- Optimization
- Python
- CVXPY

Robot Navigation Research

Researching robot control algorithms under Professor Yash Pant, learning and applying model predictive control and optimization methods.

- Model predictive control
- Optimization
- Robot control

LeetCode Practice

Solved 440+ problems, with a recent focus on design problems.

- Algorithms
- Data structures
- System design practice

Skills

Areas

Tools and Languages

Interests

Engineering